

Report

Introduction

This project uses the following dataset: <https://www.kaggle.com/datasets/udaymalviya/bank-loan-data/> data. Each model developed will attempt to predict a borrower's loan status (i.e., whether a loan is fully paid or default) based on financial and demographic predictors, and this is thus a binary classification problem. The goal of the project is to:

- Compare the performance of multiple classification models in terms of predictive accuracy, interpretability and computational cost.
- Find the most important variables for loan_status and compare the demographic and financial predictors.

Data

Data wrangling

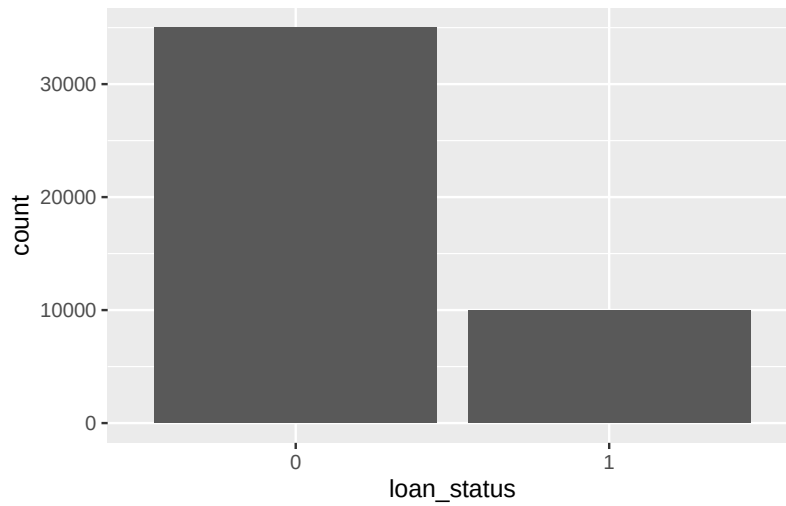
The dataset was without any NA-s, most importantly there were no NA-s for the loan_status. It did however have some potential logical outliers. We identified and removed observations that satisfied at least one of these conditions:

```
$
                                person_age < 18
                                person_age > 100
                                person_income < 0
                                loan_amnt ≤ 0
                                loan_int_rate ≤ 0
                                loan_percent_income < 0
                                person_emp_exp < 0
                                person_emp_exp > person_age
                                cb_person_cred_hist_length < 0
                                cb_person_cred_hist_length > person_age
$
```

We transformed the datatype of each column that was supposed to be a factorial to the factorial datatype, as this was processed as a character and integer in R. In addition we added the predictors emp_exp_to_age and cred_hist_to_age, which is employee experience to age ratio and credit history to age ratio respectively. These predictors might also be relevant for the models.

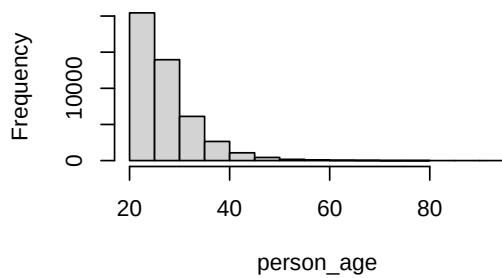
Lastly we created the train and test split. Since the default to non-default ratio is skewed (prevalence is low as seen in data analysis), we chose to stratify the splits.

Data analysis

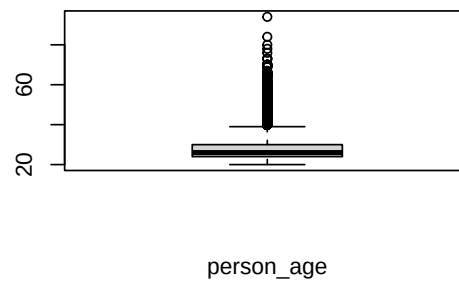


Prevalance: 0.2222568

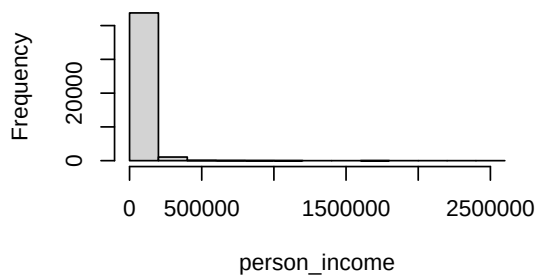
Hist, person_age



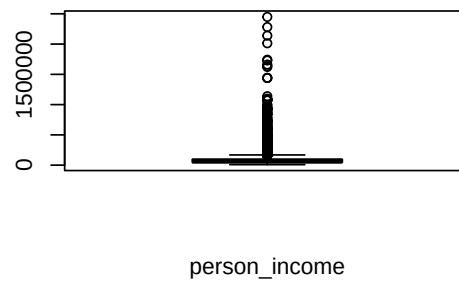
Box, person_age

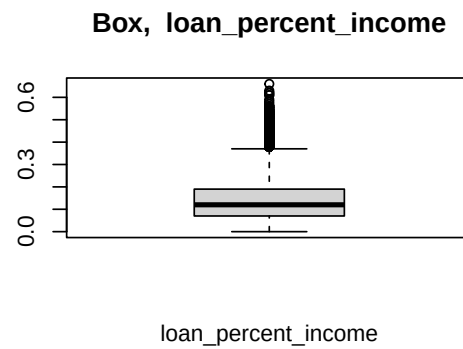
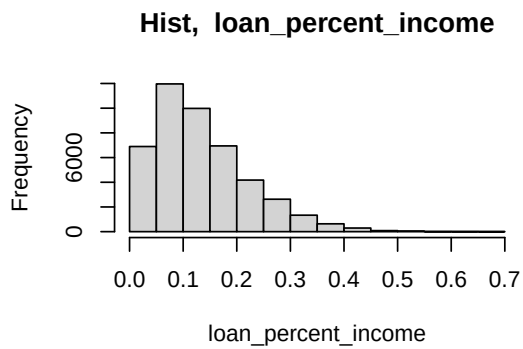
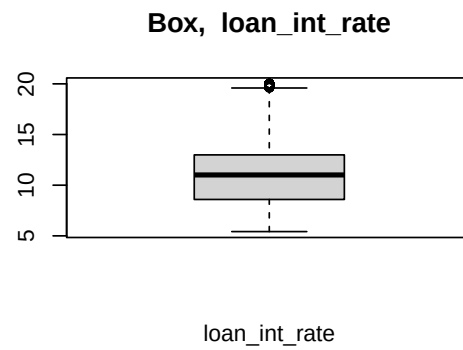
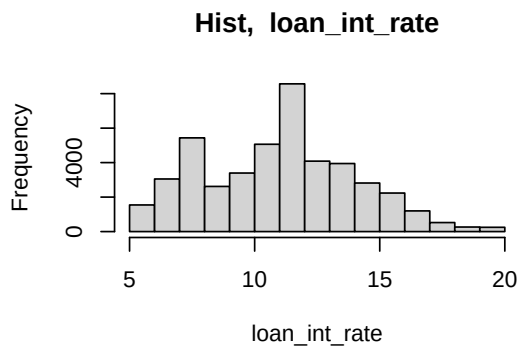
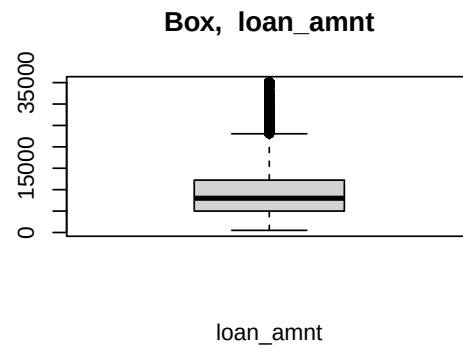
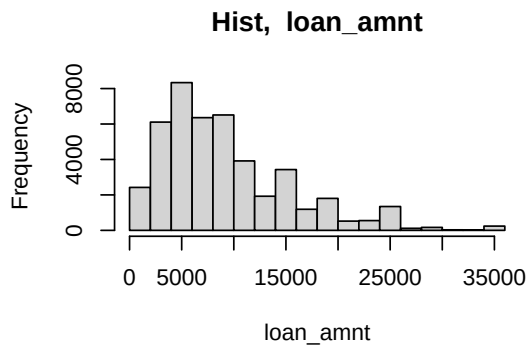
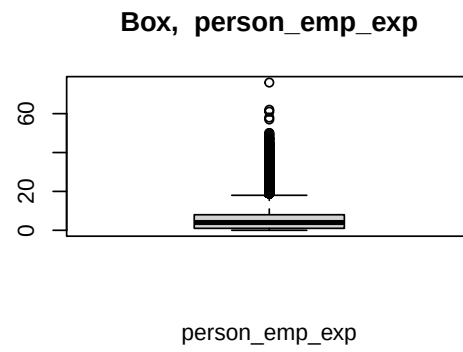
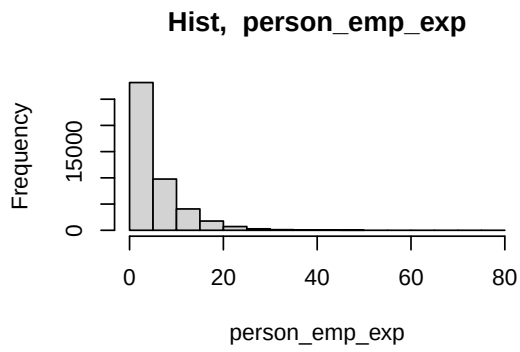


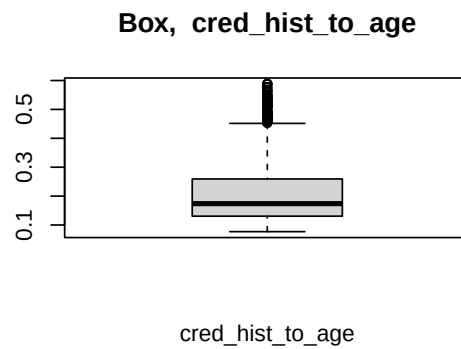
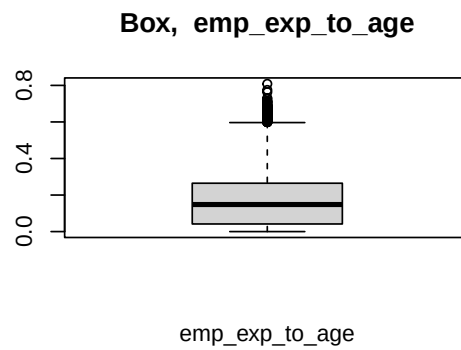
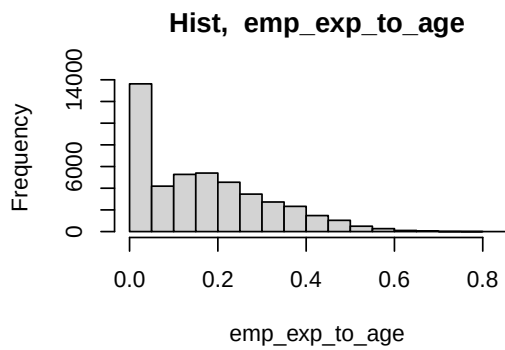
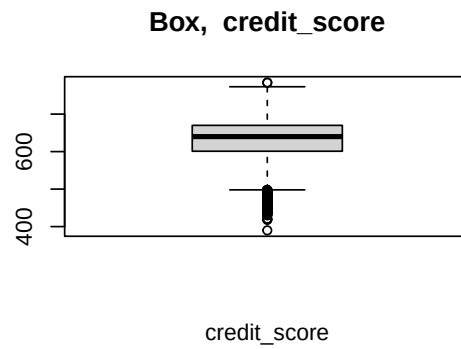
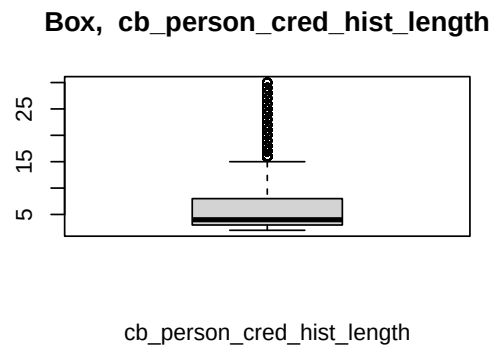
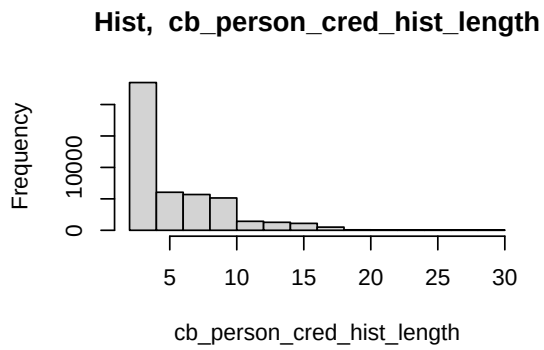
Hist, person_income

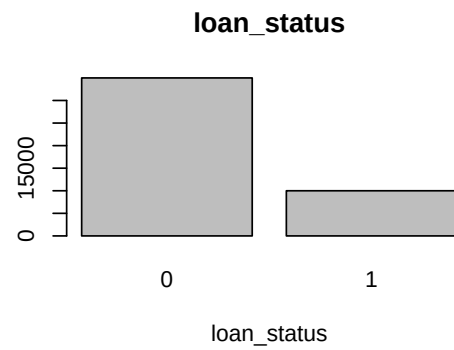
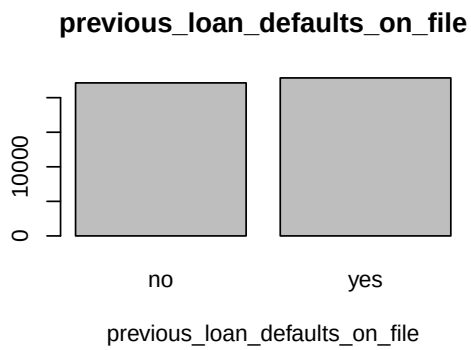
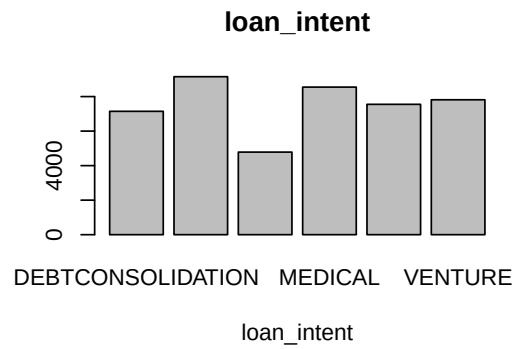
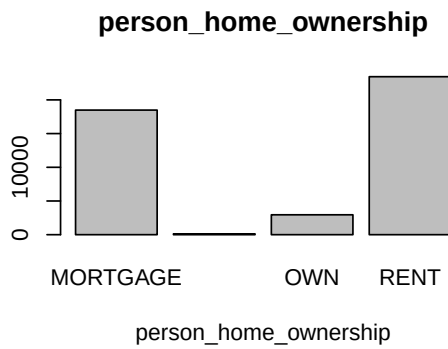
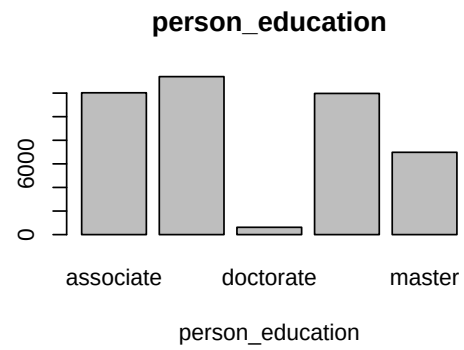
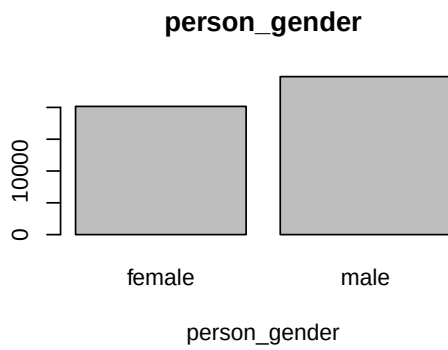


Box, person_income





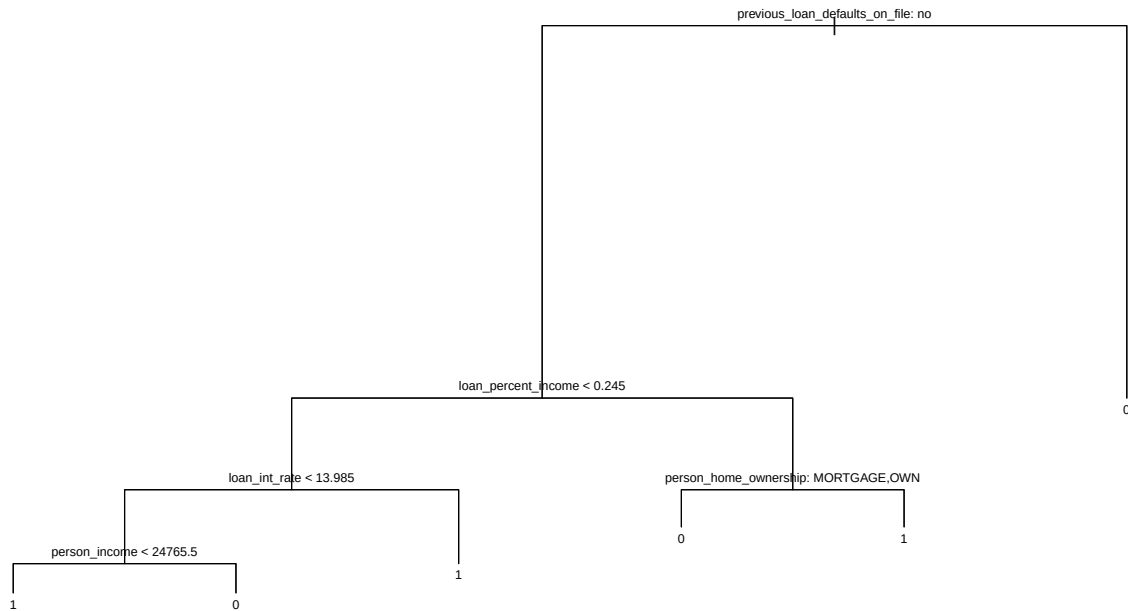




Models

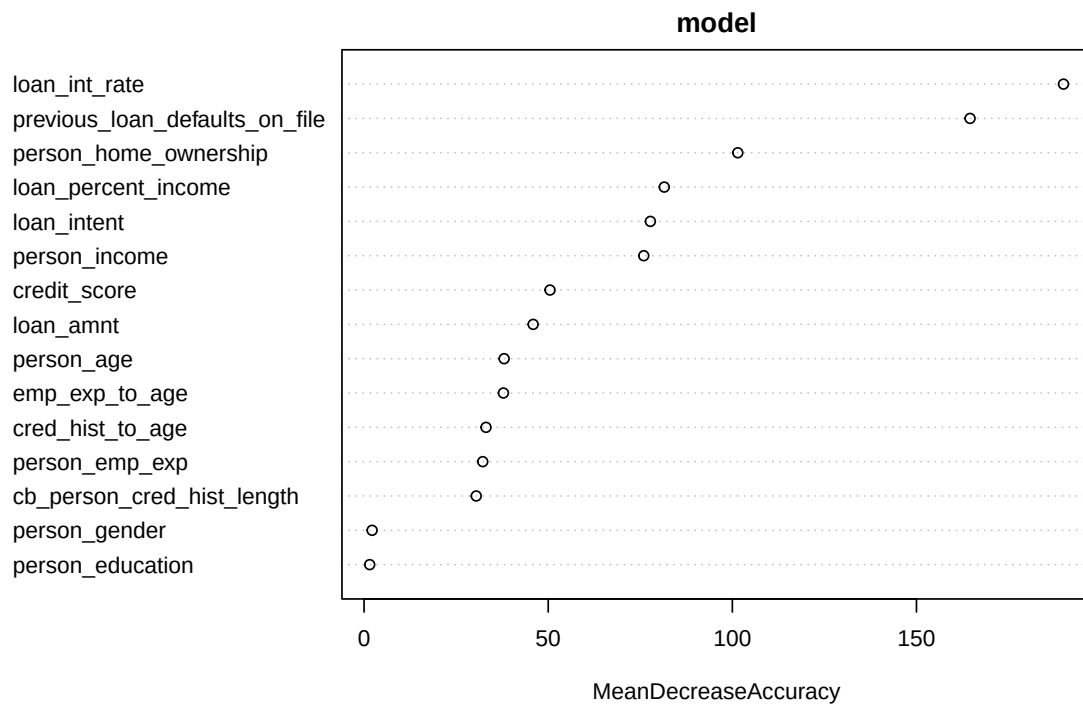
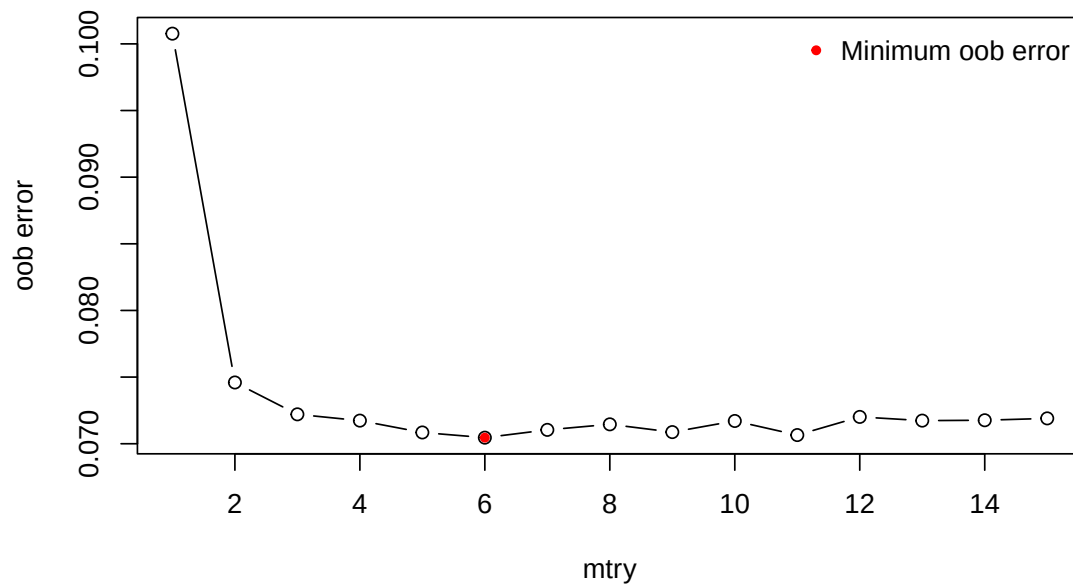
Decision trees

Single tree



```
## Misclassification rate: 0.09634404
## Sensitivity: 0.696
## Specificity : 0.9629947
## AUC: 0.9402712
```

Random forest



```
## Misclassification rate: 0.07223025
## Sensitivity: 0.766
## Specificity : 0.9739963
## AUC: 0.9742084
```

We check positives ratio since this is baseline AUC for PR. The baseline ratio is

```
## Prevalance: 0.2222469
```

Boosting

Ada Boost

```
##   iter maxdepth  nu      type      loss mean_auc
## 1   50          1 0.1 discrete exponential      NaN
```

```
## Call:
```

```
## ada(loan_status ~ ., data = train, iter = best$iter, nu = best$nu,
##      type = best$type, loss = best$loss, control = rpart::rpart.control(maxdepth = best$maxdepth,
##      cp = 0))
##
```

```
## Loss: exponential Method: discrete  Iteration: 50
```

```
##
```

```
## Training Results
```

```
##
```

```
## Accuracy: 0.851 Kappa: 0.465
```

```
##
```

```
## NULL
```

```
## Misclassification rate: 0.1479053
```

```
## Sensitivity: 0.396
```

```
## Specificity : 0.9824261
```

```
## AUC: 0.8857568
```

lgbm

```
## $objective
```

```
## [1] "binary"
```

```
##
```

```
## $metric
```

```
## [1] "average_precision"
```

```
##
```

```
## $is_unbalance
```

```
## [1] TRUE
```

```
##
```

```
## $num_leaves
```

```
## [1] 31
```

```
##
```

```
## $min_data_in_leaf
```

```
## [1] 25
```

```
##
```

```
## $learning_rate
```

```
## [1] 0.05
```

```
##
```

```
## $feature_fraction
```

```
## [1] 0.8
```

```
##
```

```
## $bagging_fraction
```



```

## [1] 0.8
##
## $bagging_freq
## [1] 1
##
## $feature_pre_filter
## [1] FALSE

## LightGBM Model (538 trees)
## Objective: binary
## Fitted to dataset with 15 columns
## LightGBM Model (538 trees)
## Objective: binary
## Fitted to dataset with 15 columns

## Misclassification rate: 0.0791199
## Sensitivity: 0.8965
## Specificity : 0.9278468
## AUC: 0.9785075

## Deviation in bootstrap:
##

##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
## 1.878e-06 3.130e-05 1.545e-04 2.975e-02 5.128e-02 3.011e-01

## Misclassification rate: 0.07511946
## Sensitivity: 0.882
## Specificity : 0.9371339
## AUC: 0.9786177

```

Tree summary

```

## AUC:

##      model      auc
## 5      bag lgbm 0.9786177
## 4      lgbm 0.9785075
## 2 random forest 0.9742084
## 1      prune 0.9402712
## 3      ada boost 0.8857568

## Misclassification rate:

##      model misclass_rate
## 3      ada boost 0.14790532
## 1      prune 0.09634404
## 4      lgbm 0.07911990
## 5      bag lgbm 0.07511946
## 2 random forest 0.07223025

```

Sensitivity:

##	model	sensitivity
## 4	lgbm	0.8965
## 5	bag lgbm	0.8820
## 2	random forest	0.7660
## 1	prune	0.6960
## 3	ada boost	0.3960

Specificity:

##	model	specificity
## 3	ada boost	0.9824261
## 2	random forest	0.9739963
## 1	prune	0.9629947
## 5	bag lgbm	0.9371339
## 4	lgbm	0.9278468